

PAF: Perturbation-Aware Filtering for Open-Set Semi-Supervised Learning

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Abstract

Open-set semi-supervised learning (OSSL) has achieved notable progress in exploiting unlabeled data, yet most existing methods overlook the distinct sensitivities of in-distribution (ID) and out-of-distribution (OOD) samples to semantic-preserving perturbations, resulting in suboptimal model performance. To address this limitation, we propose Perturbation-Aware Filtering (PAF), which leverages the behavioral difference between ID and OOD samples under perturbations and extends it into a representation-level signal for reliable OOD filtering. Specifically, PAF identifies OOD samples by measuring the representation instability under semantic-preserving perturbations. We then integrate PAF into a carefully designed two-stage training framework, allowing the model to exploit abundant unlabeled data in the first stage and gradually adapt to the open-set setting with limited labels in the second stage. Extensive experimental results on widely-used OSSL benchmarks demonstrate that our proposed PAF approach achieves superior performance compared to state-of-the-art (SOTA) OSSL methods. Our code is available at <https://github.com/njustkmg/CVPR26-PAF>.

1. Introduction

Semi-supervised learning (SSL) leverages both labeled and unlabeled data to alleviate the labeling cost while achieving comparable performance to fully supervised methods [6, 31, 50]. Traditional SSL techniques typically assume that the labeled and unlabeled datasets share identical class distributions [31]. However, such assumptions rarely hold in practice, as real-world unlabeled datasets frequently contain samples belonging to previously unseen classes [23], termed out-of-distribution (OOD) samples [40]. The presence of OOD data within unlabeled datasets severely hinders the reliability of pseudo-label generation and subsequently disrupts the model’s decision boundary [49].

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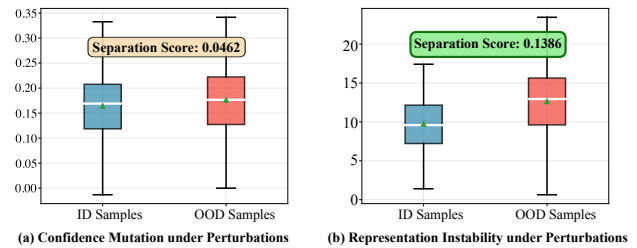


Figure 1. Comparison of (a) confidence mutation and (b) representation instability between ID and OOD samples under semantic-preserving perturbations. The separation score measures the normalized difference between ID and OOD sample distributions. It is computed as the absolute mean gap divided by the pooled standard deviation, where larger values indicate stronger distributional separability.

Consequently, accurately distinguishing in-distribution (ID) data from OOD data becomes a critical and challenging task in open-set semi-supervised learning (OSSL) [10].

Despite the progress of OSSL, effectively identifying OOD samples from unlabeled data remains a significant challenge. Recent approaches have proposed various strategies to tackle this problem. Some methods combine consistency regularization with one-vs-all decision boundaries to separate OOD and ID samples [29]. Others augment the label space with an explicit “unknown” class, allowing OOD inputs to be absorbed by the extra category [17, 25]. Meanwhile, uncertainty-based approaches further leverage evidential or Bayesian modeling to detect OOD data by assigning low-confidence samples to the unknown set [16, 45]. These methods have demonstrated efficacy under constrained settings, yet they suffer from an inherent limitation: they fail to exploit the distinct sensitivities of ID and OOD samples when the input is perturbed.

Recent literature [48] introduces a phenomenon of confidence mutation, showing that the maximum softmax probability of the OOD samples significantly fluctuates under semantic-preserving perturbations, while the ID samples remain relatively stable. However, directly measuring con-

confidence level changes alone may not fully capture subtle representation variations, as confidence is essentially a shallow reflection of deeper representation instability. A small representation change guarantees a small confidence change, but the reverse is not guaranteed. To gain deeper insights into this problem, we conduct a preliminary experiment by examining the behavior of model predictions and internal representations under different input perturbations. As shown in Figure 1, confidence mutation yields weaker ID/OOD separability, as reflected in a lower separation score. In contrast, representation instability produces a clearer separation between ID and OOD samples, achieving a much higher separation score. These results indicate that representation instability serves as a stronger and more discriminative signal for OOD detection.

Inspired by this empirical observation, we propose a novel and dynamic filtering strategy, termed Perturbation-Aware Filtering (PAF), which exploits representation-level instability under diverse perturbations to reliably separate ID from OOD samples. Furthermore, we theoretically analyze the underlying cause of the distinct sensitivities in ID and OOD samples, and demonstrate that representation-level instability provides a stronger filtering signal than confidence mutation.

Our training pipeline comprises two stages. In the first stage, we explicitly strengthen the semantic stability of labeled data representations against common perturbations by enforcing representation consistency across multiple augmented views. This consistency regularization encourages the model to preserve essential semantic information under input perturbations, a critical property leveraged by our subsequent PAF strategy. In the second stage, we deploy the PAF mechanism to dynamically filter OOD samples from the unlabeled data based on measured representation instability. Only samples exhibiting low perturbation-induced variations are retained to be jointly trained with labeled data. Extensive experimental evaluations across standard OSSSL benchmarks demonstrate the effectiveness of our proposed method. In summary, our main contributions are as follows:

- We propose a novel Perturbation-Aware Filtering (PAF) approach, which dynamically identifies OOD samples by measuring representation instability under semantic-preserving perturbations.
- We design a two-stage training framework to tailor PAF to the OSSSL scenario, which effectively incorporates limited labeled data and abundant unlabeled data, enhancing the efficacy of PAF.
- We provide a theoretical justification of the sensitivity gap between ID and OOD samples under perturbations, characterizing its relationship with the decision margin from a Lipschitz-margin perspective.
- Comprehensive empirical results demonstrate that our ap-

proach achieves superior performance over existing state-of-the-art methods on widely-used OSSSL benchmarks.

2. Related Work

2.1. Semi-Supervised Learning

Semi-supervised learning leverages abundant unlabeled data to enhance model performance, especially when labeled data are limited. Recent advances mainly fall into two categories: consistency regularization and pseudo-labeling. Early consistency-based methods [19, 26, 30, 32] enforce prediction stability under input perturbations or across teacher-student networks. Later works further enhanced consistency regularization using strong augmentations [39], MixUp strategies [2, 3], and simplified pseudo-label consistency techniques [31]. These advancements have also been extended to related tasks such as domain adaptation [4] and similarity graph matching [50]. Meanwhile, pseudo-labeling methods evolved from basic self-training [21] to adaptive strategies that refine label quality, including adaptive thresholding [13, 36, 47] and confidence-weighted soft labels [6]. Despite remarkable closed-set progress, these methods implicitly assume all unlabeled samples belong to known classes. Our work addresses this limitation by explicitly considering an open-set scenario.

2.2. Open-set Semi-Supervised Learning

Open-set semi-supervised learning tackles a more realistic setting in which unlabeled data inevitably include OOD samples from unknown classes that are absent from the labeled set [1, 22, 28, 41, 42]. Early methods in OSSSL, such as MTCF [44], propose a multi-task curriculum framework that alternately refines sample-wise OOD scores and selects likely ID unlabeled samples for training. However, these methods still have clear limitations, especially in the accuracy of OOD detection. To overcome these issues, recent studies have developed more advanced techniques. OpenMatch [29] employs a consistency-based One-vs-All (OVA) detection framework that learns decision boundaries to dynamically classify unlabeled samples as ID or OOD. IOMatch [23] fuses the predictions of a closed-set classifier and a multi-binary classifier to construct unified open-set targets, where all OOD samples are treated as a single additional class for open-set pseudo-labeling. Despite these advances, current OSSSL methods remain somewhat static in nature—typically relying on confidence derived from single-view predictions, failing to exploit instability signals under perturbations.

3. Method

3.1. Preliminaries

Problem Setting. OSSSL leverages a limited labeled dataset $\mathcal{D}_l = \{(\mathbf{x}_i^l, y_i^l)\}_{i=1}^N$ along with a large unlabeled dataset

$\mathcal{D}_u = \{\mathbf{x}_j^u\}_{j=1}^M$ collected from open-world environments, where typically $M \gg N$. Here, $\mathbf{x}_i^l$ and $\mathbf{x}_i^u$ represent labeled and unlabeled samples, and y_i^l denotes the label of $\mathbf{x}_i^l$. Unlike conventional SSL, unlabeled data inherently contain both ID samples and OOD samples. Formally, we assume an unknown partitioning of unlabeled data into $\mathcal{D}_u = \mathcal{D}_u^{\text{ID}} \cup \mathcal{D}_u^{\text{OOD}}$, where samples from $\mathcal{D}_u^{\text{ID}}$ share the same distribution as the labeled set $\mathcal{D}_l$, while $\mathcal{D}_u^{\text{OOD}}$ consists of samples from unknown categories. The primary objective of OSSL is to simultaneously learn a robust classifier capable of accurately classifying ID samples and effectively identifying and filtering out OOD samples to prevent them from adversely affecting the learned decision boundaries.

Approach Overview. Our approach addresses the OSSL challenge via a two-stage framework that exploits the inherent sensitivity gap between ID and OOD samples under perturbations. In the first stage, we strengthen the representation stability of ID samples under perturbations by employing a representation consistency regularization. Considering the limitation of labeled data, we simultaneously introduce an auxiliary self-supervised task, to enrich the semantic representation capability of the WideResNet backbone [46]. In the second stage, we integrate Perturbation-Aware Filtering (PAF) to dynamically partition the unlabeled dataset into ID and OOD subsets based on the representation instability under diverse perturbations. For unlabeled data with stable representations, we further employ an instability-weighted pseudo-labeling mechanism, which leverages perturbation stability to regulate the contribution of pseudo-ID samples during training. To prevent task interference across learning objectives, we assign dedicated projection modules and classification heads (hereafter referred to as projection heads for simplicity) to the self-supervised and consistency branches, ensuring independent gradient flows. Figure 2(a) summarizes the proposed two-stage training framework, while Figure 2(b) shows the PAF procedure to identify OOD samples from unlabeled data.

3.2. Perturbation-Aware Filtering

Given an unlabeled sample $\mathbf{x}^u$, we apply semantic-preserving perturbation operator $\mathcal{A}$ (e.g., weak augmentations such as flipping and mild color jitter) to generate K_a perturbed views, while ensuring the image semantics remain intact. Let $\varphi(\cdot)$ denote the backbone representation extractor followed by the consistency projection head $\phi_{\text{con}}(\cdot)$. We then compute the corresponding penultimate-layer representations for these perturbed views. Under these semantic-preserving perturbations, the representation-level instability $R(\cdot)$ can be quantified as follows:

$$R(\mathbf{x}^u) = \frac{1}{K_a d} \sum_{k=1}^{K_a} \left\| \phi_{\text{con}}(\varphi(\mathbf{x}^u)) - \phi_{\text{con}}(\varphi(\mathbf{x}_{(k)}^u)) \right\|_2^2, \quad (1)$$

where d is the representation dimension and $\mathbf{x}_{(k)}^u$ denotes the k -th perturbed view.

In practice, we adopt $K_a = 1$ for efficiency. Next, we dynamically determine an adaptive threshold τ using an Otsu-based criterion [27] that automatically adapts to the batch-wise distribution of representation-level instability. Formally, we partition the unlabeled set as:

$$\mathcal{X}_u^{\text{ID}} = \{\mathbf{x}^u \mid R(\mathbf{x}^u) < \tau\}, \quad \mathcal{X}_u^{\text{OOD}} = \{\mathbf{x}^u \mid R(\mathbf{x}^u) \geq \tau\}. \quad (2)$$

3.3. Two-Stage Framework

Representation Consistency Regularization. Since our PAF strategy distinguishes ID and OOD samples by measuring the instability of the representation under perturbations, it is crucial to learn perturbation-invariant representations. To achieve this, we explicitly impose a semantic consistency constraint on the representations extracted by our backbone. The same loss is first applied to the labeled data during the first stage and later to the filtered pseudo-ID samples $\mathcal{X}_u^{\text{ID}}$ in the second stage.

Given an input image $\mathbf{x}$, we generate two weakly augmented views as $\mathbf{x}' = \mathcal{A}_1(\mathbf{x})$ and $\mathbf{x}'' = \mathcal{A}_2(\mathbf{x})$, respectively. This perturbation pairing enables us to enforce consistency in the representation space.

We use a consistency projection head $\phi_{\text{con}}(\cdot)$ to map the backbone representation into a compact embedding space optimized for stability across these perturbations. Formally, the consistency loss is defined as the mean squared difference between the embeddings of the two augmented views:

$$\mathcal{L}_{\text{con}} = \frac{1}{d|\mathcal{D}_l|} \sum_{\mathbf{x} \in \mathcal{D}_l} \left\| \phi_{\text{con}}(\varphi(\mathbf{x}')) - \phi_{\text{con}}(\varphi(\mathbf{x}'')) \right\|_2^2, \quad (3)$$

where d is the representation dimension.

By explicitly minimizing this embedding discrepancy, we encourage the learned representations to remain stable under semantic-preserving perturbations. This perturbation-invariant representation space significantly enhances the performance of our approach.

Self-Supervised Semantic Structuring. Our PAF approach fundamentally relies on a robust representation extraction backbone capable of distinguishing subtle semantic shifts. However, due to limited labeled data in the early stage of training, it is challenging for the backbone to learn sufficiently discriminative representations. To overcome this limitation, we introduce an auxiliary self-supervised rotation prediction task [9], which encourages the backbone to learn richer semantic structures. The same loss is applied to all samples during both stages.

Specifically, each sample is randomly rotated using: $\pi = \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$, and the model is trained to identify

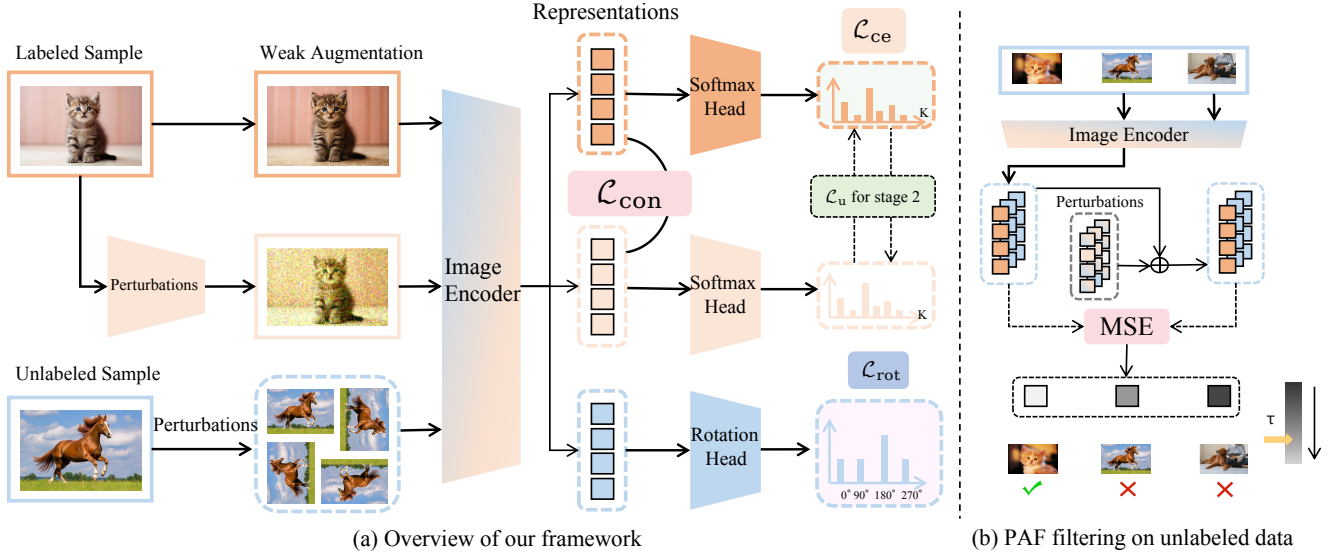


Figure 2. (a) Overview of our framework, the model learns semantic representations via four complementary components: supervised classification on labeled data, rotation prediction to help model improve semantic structure, representation consistency regularization to enforce model to learn perturbation-invariant representation, and an instability-weighted pseudo-labeling mechanism, where samples are adaptively weighted based on their representation instability. (b) PAF filtering on unlabeled data, we measure the representation instability under perturbations to estimate sample reliability. An Otsu-based threshold τ dynamically filters out potential OOD samples from the unlabeled set. Only samples with stable representation are retained for subsequent training.

the applied rotation angle, which inherently requires understanding image content at a higher semantic level. This process is formalized as minimizing a cross-entropy loss:

$$\mathcal{L}_{rot} = -\frac{1}{4|\mathcal{D}_l \cup \mathcal{D}_u|} \sum_{\mathbf{x} \in \mathcal{D}_l \cup \mathcal{D}_u} \sum_{r \in \pi} \log p(r | \phi_{rot}(\mathbf{x}^r)), \quad (4)$$

where ϕ_{rot} represents the dedicated rotation projection head, $\mathbf{x}^r$ denotes an image rotated by angle r , and $p(r | \cdot)$ is the predicted probability for the rotation class r .

Instability-Weighted Pseudo-Labeling. To fully exploit the ID subset $\mathcal{X}_u^{ID}$ returned by PAF, we design an instability-weighted pseudo-labeling loss that extends UDA [39]. Unlike vanilla UDA, which treats all unlabeled samples equally, our loss incorporates the perturbation-induced instability $\delta(\mathbf{x}^u)$ of each sample to modulate its contribution.

For every $\mathbf{x}^u \in \mathcal{X}_u^{ID}$, we generate a weakly augmented view $\mathbf{x}^w$ and a strongly augmented view $\mathbf{x}^s$. Let p^w and p^s be their softmax predictions, and let δ be the representation gap between $\mathbf{x}^w$ and $\mathbf{x}^s$ defined by the ℓ_2 term in Eq. (3). We first construct a pseudo-label from p^w if its maximum probability exceeds the confidence threshold α . Then we minimize a weighted KL divergence:

$$\mathcal{L}_u = \frac{1}{|\mathcal{X}_u^{ID}|} \sum_{\mathbf{x}^u \in \mathcal{X}_u^{ID}} \underbrace{(1 - \hat{\delta}(\mathbf{x}^u))}_{\text{instability-aware weight}} \times \mathbb{1}[\max(p^w) \geq \alpha] \text{KL}(p^w \| p^s), \quad (5)$$

where $\hat{\delta}(\mathbf{x}^u)$ is the min-max normalized instability score so that more stable samples ($\hat{\delta} \rightarrow 0$) receive larger weights, and $\mathbb{1}[\cdot]$ denotes the indicator function. This design tightly couples pseudo-label learning with our representation consistency criterion: samples that are both confident and stable drive the decision boundary, while less stable ones are automatically down-weighted or discarded. The loss is activated in the second stage, where it is applied to the pseudo-ID samples filtered by PAF in Sec. 3.2.

3.4. Overall Training Process

In summary, our training process consists of two stages (shown in Algorithm (1)). In the first stage, we perform warm-up training to optimize the entire architecture using the loss:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{con} + \mathcal{L}_{rot}, \quad (6)$$

where $\mathcal{L}_{ce}$ is the standard cross-entropy loss for labeled samples. In the second stage, we incorporate our PAF strategy to separate ID and OOD samples. We periodically filter out reliable ID samples and jointly train them with the labeled data. The loss function in the second stage is:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_u + \mathcal{L}_{con} + \mathcal{L}_{rot}. \quad (7)$$

4. Theoretical Analysis

In this section, we present the theoretical basis of our PAF strategy. We first establish in Sec. 4.1 that the model's re-

Algorithm 1 Training process of our framework

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1: Input: Labeled set  $\mathcal{D}_l$ , unlabeled set  $\mathcal{D}_u$ , number of
   stage 1 epochs  $E_1$ , number of stage 2 epochs  $E_2$ .
2: Stage 1:
3: for  $e = 1$  to  $E_1$  do
4:   Compute  $\mathcal{L}_{ce}$  on  $\mathcal{D}_l$ .
5:   Compute  $\mathcal{L}_{con}$  on  $\mathcal{D}_l$  based on Eq. (3).
6:   Compute  $\mathcal{L}_{rot}$  on all samples based on Eq. (4).
7:   Update model using  $\mathcal{L}_{ce} + \mathcal{L}_{con} + \mathcal{L}_{rot}$ .
8: end for
9: Stage 2:
10: Load checkpoint from Stage 1.
11: for  $e = 1$  to  $E_2$  do
12:   if  $e \bmod 20 = 0$  then
13:     Filter ID samples from  $\mathcal{D}_u$  via PAF strategy.
14:   end if
15:   Compute  $\mathcal{L}_{ce}$  on  $\mathcal{D}_l$ .
16:   Compute  $\mathcal{L}_u$  on  $\mathcal{X}_u^{ID}$  based on Eq. (5).
17:   Compute  $\mathcal{L}_{con}$  on  $\mathcal{X}_u^{ID}$ .
18:   Compute  $\mathcal{L}_{rot}$  on all samples.
19:   Update model using  $\mathcal{L}_{ce} + \mathcal{L}_u + \mathcal{L}_{con} + \mathcal{L}_{rot}$ .
20: end for
21: Output: Model parameter  $\theta$ .
```

sponse to semantic-preserving perturbations becomes exponentially less sensitive as the decision margin increases. Building upon this, Sec. 4.2 further extends the analysis to the representation space, linking confidence fluctuation to representation variation and clarifying how perturbation sensitivity is governed by representation dynamics.

4.1. A Lipschitz-Margin View of Perturbation Sensitivity

Let $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^K$ be the logit map and $\sigma(\cdot)$ the softmax. For an input $\mathbf{x}$, define:

$$s(\mathbf{x}) = \max_k \sigma_k(f_\theta(\mathbf{x})), \quad y(\mathbf{x}) = \arg \max_k f_{\theta,k}(\mathbf{x}).$$

where k indexes the class dimension and $s(\mathbf{x})$ denotes the predicted class confidence. We further define the decision margin:

$$m(\mathbf{x}) = f_{\theta,y(\mathbf{x})}(\mathbf{x}) - \max_{k \neq y(\mathbf{x})} f_{\theta,k}(\mathbf{x}).$$

Assumption. Each logit $f_{\theta,k}$ is β -Lipschitz [33]. Our empirical measurements in Supplementary Sec. A confirm that the assumption remains reasonable in practice.

Lemma 1 (Gradient-margin bound). *For any $\mathbf{x}$ with margin $m(\mathbf{x}) > 0$, the gradient of the top-confidence score satisfies*

$$\|\nabla_{\mathbf{x}} s(\mathbf{x})\|_2 \leq \beta (K-1) e^{-m(\mathbf{x})}.$$

The proof of Lemma 1 is provided in Supplementary Sec. B. Building on this result, we further establish the following variance bound:

Theorem 1 (Variance upper bound). *Draw an isotropic perturbation $\delta \sim \mathcal{P}$ with $\mathbb{E}[\delta] = \mathbf{0}$ and $\mathbb{E}[\|\delta\|_2^2] = \sigma^2$, and assume $\|\delta\|_2 \leq \xi$ almost surely. Then for any $\mathbf{x}$,*

$$\text{Var}_\delta[s(\mathbf{x} + \delta)] \leq \beta^2 (K-1)^2 e^{-2m(\mathbf{x})} \sigma^2.$$

Theorem 1 implies that perturbation-induced prediction variance decays exponentially as the decision margin $m(\mathbf{x})$ increases. ID samples with large $m(\mathbf{x})$ are intrinsically stable, whereas OOD or boundary-close samples with small margins remain highly sensitive to perturbations. This provides a principled explanation for why OOD samples exhibit higher fluctuation than ID samples under semantic-preserving perturbations. The proof is provided in the Supplementary Sec. B.

4.2. Representation Confidence Coupling

To further connect confidence stability with internal representations, we reformulate the gradient bound in terms of the penultimate representation space.

Proposition 1. *Let $\varphi(\mathbf{x}) \in \mathbb{R}^d$ be the penultimate layer representation, and let the classifier be $f_\theta(\mathbf{x}) = W\varphi(\mathbf{x}) + b$. Here we denote $\mathbf{x}'$ as a semantics-preserving perturbed view of $\mathbf{x}$. Because each logit is β -Lipschitz in φ with $\beta := \|W\|_2$, the previous gradient bound can be rewritten with respect to representation variation:*

$$|s(\mathbf{x}) - s(\mathbf{x}')| \leq \beta (K-1) e^{-m(\mathbf{x})} \|\varphi(\mathbf{x}) - \varphi(\mathbf{x}')\|_2.$$

Hence a small representation change guarantees a small confidence change, but the converse need not hold. Detecting representation instability $\|\varphi(\mathbf{x}) - \varphi(\mathbf{x}')\|_2$ is therefore a stricter criterion, which supports our representation-level filter. The proof is provided in Supplementary Sec. B.

5. Experiments

5.1. Experimental Setup

Datasets. We evaluated our approach in the following two complementary settings: internal OOD scenario and external OOD scenario. For internal OOD scenario, following Ma et al. [25], we draw OOD samples from the same dataset but from categories not seen during training. We fix the number of labeled and unlabeled images and vary the mismatch ratios, defined as the proportion of OOD samples in the unlabeled data. For example, a mismatch ratio of 0.3 means 30% of the unlabeled data are OOD samples. In detail, we choose six ID classes ('0'-'5') with ten labeled samples per class and 30,000 unlabeled samples for MNIST [20]; six ID classes with 400 labeled samples per

Table 1. Seen-class classification accuracy (%) of different methods on four datasets.

Method	MNIST		CIFAR-10		CIFAR-100		TinyImageNet	
	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
PL	90.0±0.7	86.0±0.6	74.6±0.7	75.8±0.8	60.2±0.3	57.5±0.6	36.6±0.6	35.8±0.4
Pi-Model	92.4±0.6	86.6±0.5	75.7±0.7	74.5±1.0	59.4±0.3	57.9±0.3	36.9±0.4	36.4±0.5
VAT	94.5±0.3	90.4±0.3	75.0±0.5	80.5±0.4	61.8±0.4	59.6±0.6	36.7±0.5	36.3±0.6
FixMatch		–	81.5±0.2	80.9±0.3	65.9±0.3	65.2±0.3		–
DS ³ L	96.8±0.3	94.5±0.4	78.1±0.4	76.9±0.5		–		–
UASD	96.2±0.6	94.3±0.8	77.6±0.4	77.6±0.4	61.8±0.4	58.4±0.5	37.1±0.7	36.9±0.6
CL	96.9±0.1	95.6±0.4	82.1±0.4	82.1±0.5	63.6±0.4	61.5±0.5	37.3±0.7	37.0±0.5
OpenMatch	97.8±0.2	96.0±0.2	88.2±0.2	85.5±0.5	68.7±0.1	68.4±0.2	37.9±0.4	37.0±0.3
T2T	94.8±0.1	96.2±0.2	89.0±0.4	86.9±0.2	69.8±0.2	68.0±0.2	39.1±0.3	37.3±0.4
SAFE-STUDENT	93.0±0.3	96.5±1.0	83.8±0.3	86.2±0.5	69.0±0.3	68.3±0.2	37.7±0.3	37.1±0.3
Ma <i>et al.</i>	98.7±0.2	96.9±0.1	91.4±0.3	89.1±0.1	72.5±0.2	70.4±0.1	40.8±0.3	39.9±0.3
ANEDL	98.9±0.2	98.6±0.3	91.8±0.4	90.1±0.2	75.4±0.3	73.8±0.4	47.6±0.5	45.8±0.5
SCOMatch	99.0±0.1	99.0±0.1	92.2±0.2	90.2±0.6	75.3±0.5	73.5±0.2	54.2±0.3	52.3±0.3
BDMatch	99.1±0.1	99.1±0.1	92.5±0.3	90.4±0.4	75.9±0.5	73.7±0.3	56.7±0.3	52.9±0.3
CaliMatch	98.8±0.2	97.6±0.2	90.3±0.3	87.6±0.4	72.3±0.3	68.6±1.0	47.5±0.7	44.7±0.4
Ours	99.4±0.1	99.2±0.1	93.1±0.6	90.8±0.6	77.7±0.6	75.9±0.4	58.1±0.3	54.6±0.3

Table 2. AUC (%) for OOD class identification of different methods on four datasets.

Method	MNIST		CIFAR-10		CIFAR-100		TinyImageNet	
	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
MTCF	95.0±0.3	91.4±0.2	64.2±0.4	59.8±0.5	61.3±0.6	56.4±0.4	37.8±0.7	34.2±0.6
T2T	90.6±0.1	91.2±0.3	67.4±0.3	62.8±0.4	64.8±0.5	60.1±0.6	40.5±0.1	37.0±0.5
OSP	97.0±0.2	93.0±0.1	88.3±1.1	83.6±0.8	71.8±0.1	65.2±0.3	54.4±0.6	50.1±0.2
ProSub	98.0±0.3	98.2±0.4	89.1±0.5	84.5±0.6	73.2±0.3	67.0±0.7	56.2±0.6	52.0±0.5
Ours	98.2±0.1	98.4±0.2	96.1±0.2	93.6±0.3	84.5±0.4	76.7±0.3	64.8±0.5	61.1±0.4

class and 20,000 unlabeled samples for CIFAR-10 [18]; the first 50 classes as ID with 100 labeled samples per class and 20,000 unlabeled samples for CIFAR-100 [18]; the first 100 classes as ID with 100 samples per class and 40,000 unlabeled samples for TinyImageNet [8]. For external OOD scenario, following the setup in T2T [15], we select all classes of CIFAR-10 as ID classes and externally introduce two real-world OOD datasets (TinyImageNet and LSUN [43]) and two synthetic OOD datasets (Gaussian Noise and Uniform Noise) [15]. Each OOD dataset contributes 10,000 images, combined with CIFAR-10 to form open-set scenarios.

Implementation Details. When training with MNIST, we employ a two-layer CNN as the backbone network using stochastic gradient descent (SGD) with a learning rate of 10^{-3} [10]. For CIFAR-10, CIFAR-100, and TinyImageNet, WideResNet28-2 [46] is used as the backbone network. We optimize the network using SGD with an initial learning rate

of 0.03 and a momentum of 0.9. We set the batch size of labeled and unlabeled samples as 64 and 128 respectively. For our method, the first stage costs 50,000 iterations, and the second stage costs 200,000 iterations. All experiments in this work are conducted over three runs with different random seeds, and we report the mean and standard deviation.

5.2. Comparison with SOTAs

To illustrate the effectiveness of our method on OSSL, we compare it against several state-of-the-art methods, including PL [21], Pi-Model [30], MixM [2], VAT [26], UDA [39], FixMatch [31], DS³L [10], UASD [7], MTCF [44], OTCT [24], CL [5], OpenMatch [29], T2T [15], SAFE-STUDENT [12], Ma *et al.* [25], OSP [37], ANEDL [45], ProSub [35], SCOMatch [38], BDMatch [11] and CaliMatch [1]. We use seen-class classification accuracy to evaluate the performance of classifier and AUC [14] to evaluate the performance of OOD detection.

Table 3. Seen-class classification accuracy (%) on CIFAR-10 under various external OOD datasets.

Method	TIN			LSUN			Gaussian Noise			Uniform Noise		
	250	1000	4000	250	1000	4000	250	1000	4000	250	1000	4000
MixM	82.4±0.7	88.0±0.2	91.3±0.1	76.3±4.1	87.0±0.4	91.3±0.4	75.7±3.5	85.7±1.1	91.5±0.4	72.90±0.9	84.5±0.9	90.5±0.4
UDA	88.8±0.5	91.8±0.1	93.6±0.1	88.5±1.1	91.1±0.3	93.3±0.1	88.9±0.7	89.2±0.4	92.4±0.2	88.7±0.9	89.7±0.4	92.7±0.4
UASD	83.5	-	-	80.9	-	-	-	-	-	-	-	-
DS ³ L	70.1±4.8	82.6±0.1	-	69.7±0.9	-	82.9±1.0	62.7±0.7	-	80.4±0.0	62.9±1.7	-	80.6±0.1
MTCF	86.4±0.6	89.9±0.1	93.0±0.4	86.7±0.4	90.2±0.5	92.9±0.3	87.3±0.1	89.8±0.3	92.5±0.1	85.5±0.1	89.9±0.1	92.8±0.0
OTCT	-	91.1±0.7	93.8±0.1	-	91.3±0.4	94.3±0.2	-	92.3±0.6	94.5±0.1	-	91.8±0.0	94.5±0.1
T2T	91.5±0.1	93.2±0.1	92.9±0.1	91.1±0.2	92.3±0.1	93.4±0.1	90.8±0.1	91.8±0.1	94.4±0.1	90.0±0.2	92.7±0.1	93.4±0.2
CaliM	91.8±0.1	93.0±0.0	93.1±0.1	91.4±0.1	92.6±0.1	93.5±0.2	91.0±0.1	92.5±0.1	94.5±0.4	90.3±0.1	93.2±0.1	93.8±0.3
Ours	92.4±0.2	93.5±0.1	93.9±0.1	91.7±0.2	93.3±0.1	93.7±0.1	91.6±0.1	93.4±0.1	94.6±0.1	91.4±0.2	93.8±0.1	94.8±0.2

Table 4. Comparison with CFSG-CLIP on the CUB-200-2011 dataset under the fine-grained OSSL setting. Following the original paper, we designate the odd classes as ID classes and the even classes as OOD classes. For each ID class, we randomly select 5 samples as labeled data. We report seen-class accuracy (Acc, %).

Method	CUB-200-2011
CFSG-CLIP	84.73
CFSG-CLIP + PAF (ours)	85.16

We evaluate our approach under the internal OOD scenario, and the results are summarized in Table 1. Our method consistently outperforms state-of-the-art approaches across all datasets, underscoring the effectiveness of the perturbation-aware filtering strategy. Table 2 further shows that PAF achieves the best AUC on all four datasets, with larger improvements on CIFAR-100 and TinyImageNet, suggesting that PAF provides a more reliable OOD filtering signal when the separation task is harder. The results on CIFAR-10 combined with external OOD datasets are summarized in Table 3. As reported in Table 3, our method achieves significant performance improvements under external OOD settings, exhibiting strong robustness even in the presence of severe distribution shifts such as Gaussian Noise and Uniform Noise.

Additionally, we integrate PAF into CFSG-CLIP [22], a recent CLIP-driven framework for fine-grained OSSL. As shown in Table 4, incorporating our approach into CFSG-CLIP consistently improves seen-class accuracy on CUB-200-2011 [34]. This demonstrates that PAF remains effective within vision-language frameworks and provides complementary cues beyond textual semantic guidance.

5.3. Ablation Study

To evaluate the effectiveness and necessity of each component in our framework, we conduct comprehensive ablation experiments on CIFAR-100 under a mismatch ratio of 0.3.

Table 5. Ablation results on CIFAR-100 under a mismatch ratio of 0.3. We report seen-class accuracy (Acc, %) and AUC (%).

Method	Acc	AUC
Full PAF (ours)	77.7	84.5
w/o perturbation-aware filtering	71.2	74.8
w/o separate projection head	61.8	70.2
w/o representation consistency loss	75.1	81.3
w/o auxiliary rotation loss	75.6	81.2
confidence-based filtering	74.5	78.2

Experimental results are summarized in Table 5.

Effect of Perturbation-Aware Filtering. As shown in Table 5, removing the PAF strategy significantly decreases the model’s seen-class accuracy. To further demonstrate the advantage of our filtering strategy, we compare it with confidence-based filtering [48], which instead relies on the stability of maximum confidence scores under input perturbations. PAF achieves consistently higher accuracy and AUC than confidence-based filtering mechanism, as shown in Table 5. We additionally evaluate the ID sample retention rate and the OOD sample filtering rate of the two filtering mechanisms. Figure 3(a) shows that our method consistently retains a higher percentage of ID samples than confidence-based filtering. Figure 3(b) shows that our method effectively removes more OOD samples, ensuring cleaner model supervision, which is consistent with our coupling analysis in Sec. 4.2.

Effect of Separate Projection Head. Removal of the separate projection head leads to a significant performance drop (nearly 16%). In our experiments, we observed that the representation consistency loss $\mathcal{L}_{\text{con}}$ and self-supervised loss $\mathcal{L}_{\text{rot}}$ produce conflicting gradients: the former aligns representations of the same image under perturbations, while the latter separates them into distinct orientation subspaces. When both objectives share a single projection space, their gradients often point in opposing directions. To mitigate

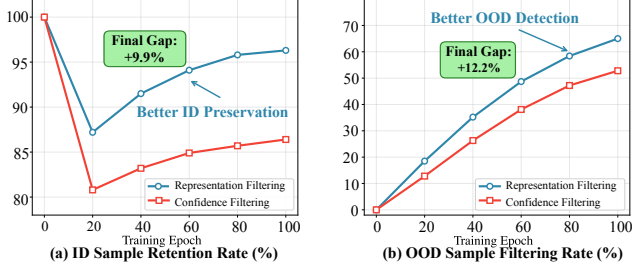


Figure 3. Comparison between representation-based and confidence-based filtering strategies on CIFAR-100 under a mismatch ratio 0.3. (a) ID sample retention rate. (b) OOD sample filtering rate.

this, we introduce two task-specific projection heads, each mapping backbone features into separate subspaces. This design prevents gradient conflict, enabling stable model learning.

Effect of Representation Consistency Loss. We further assess the contribution of the representation consistency loss. Removing $\mathcal{L}_{\text{con}}$ leads to a noticeable accuracy degradation (around 3% lower than the full method). This emphasizes the essential role of the representation consistency constraint, which explicitly enforces consistency of learned representations to input perturbations.

Effect of Auxiliary Rotation Loss. As shown in Table 5, removing $\mathcal{L}_{\text{rot}}$ causes a clear degradation in both seen-class accuracy and AUC. This auxiliary task provides an additional self-supervised signal, encouraging the backbone to capture richer semantic structures that underpin more discriminative representations.

5.4. Further Analysis

Parameter Sensitivity. We analyze the sensitivity of our method to two critical factors: (i) the number of perturbation trials K_a for estimating representation instability, and (ii) the thresholding method that determines which unlabeled samples are retained as pseudo-ID. We report seen-class accuracy (Acc, %) and AUC (%) in Figure 4.

Perturbation Trials K_a . Setting $K_a=1$ offers an effective balance between performance and efficiency. While increasing K_a can lead to a slight gain in AUC, the improvement remains marginal relative to the considerable rise in computational cost.

Thresholding Strategies. To validate the effectiveness of our Otsu-based thresholding [27], we first apply a mild rescaling factor γ and set the scaled threshold as $\tau = \gamma T_{\text{Otsu}}$ to examine whether the Otsu threshold remains stable under rescaling perturbations. Then we compare it with two alternatives: $F(x)$ denotes a fixed threshold value (e.g., $x=0.3$), and $P(p)$ represents a percentile-based threshold determined by the p th percentile of instability scores. As shown in Fig-

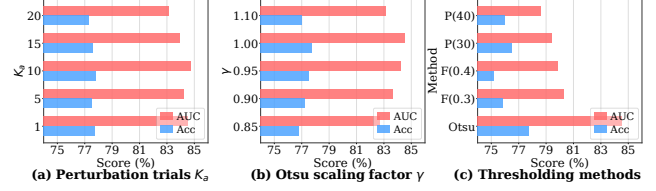


Figure 4. Sensitivity analysis on CIFAR-100 under mismatch ratio 0.3. (a) Impact of perturbation trials K_a . (b) Impact of Otsu scaling factor γ . (c) Comparison with thresholding methods: $F(x)$ denotes a fixed threshold and $P(p)$ a percentile-based threshold.

Table 6. Computational overhead of PAF. Measurements are taken on an NVIDIA RTX 4090 with batch size 64.

Method	Training time (hours)	Peak GPU memory(GB)
T2T	29.3	5.4
SCOMatch	28.6	4.3
PAF	23.1	5.4

ure 4(b), the performance of our method varies smoothly when γ changes around 1.0, confirming the robustness of the Otsu-based selection. In contrast, fixed or percentile thresholds exhibit more abrupt performance degradation, as illustrated in Figure 4(c), since they fail to adapt to the evolving score distribution during training. These results demonstrate that Otsu-based scheme provides both stability and adaptability, yielding consistently better Acc and AUC across settings.

Overhead. Table 6 reports the training time and peak GPU memory measured under identical settings. PAF shows reduced overall training time compared with T2T [15] and SCOMatch [38], with similar peak memory usage, suggesting that PAF is computationally efficient in practice.

6. Conclusion

In this paper, we propose a novel OSSL method called Perturbation-Aware Filtering (PAF), which dynamically distinguishes ID and OOD samples by measuring representation instability under semantic-preserving perturbations. We then integrate PAF into a carefully designed two-stage framework that exploits abundant unlabeled data during the first stage and activates PAF in the second stage to achieve reliable ID–OOD separation. We further provide a theoretical justification for our method. Extensive experiments on MNIST, CIFAR-10, CIFAR-100, and TinyImageNet confirm the superior performance of our approach over existing methods. We also performed ablation studies to validate the effectiveness of each component of our method, demonstrating the rationality of our overall design.

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